

SUMMARY

Moore and Mealy machines are specialized forms of Finite Automata categorized as Output Producers rather than Language Acceptors. Unlike DFA or NFA, which produce binary acceptance results (Yes/No), these machines generate output strings corresponding to input strings.

Key distinctions:

Moore machines associate output with states.

Mealy machines associate output with transitions.

Output length and response to ϵ differ significantly between the two.

Recent competitive examinations highlight their importance due to fast solvability when core principles are understood.

FUNDAMENTAL CONCEPTS OF FINITE AUTOMATA WITH OUTPUT

Finite Automata can be classified into two categories:

1. Language Acceptors (Without Output)

Models such as DFA, NFA, and ϵ -NFA determine whether an input string belongs to a language. The result is binary: Accepted or Rejected.

2. Output Producers (With Output)

Moore and Mealy machines transform an input string into an output string.

Applications include:

Binary addition

1's complement and 2's complement

Small counting operations

These machines operate without auxiliary memory beyond finite-state capability.

SHARED ARCHITECTURAL PRINCIPLES

Moore and Mealy machines inherit properties from DFA theory.

Specialized DFA Case:

Transition function must be defined for every state and input symbol.

Absence of Final States:

No final states are required since acceptance is not the objective.

Absence of Dead States:

Dead states are unnecessary because strings are not rejected.

Equivalent States:

Two states may be equivalent if they produce identical outputs under identical conditions.

DETAILED ANALYSIS OF MOORE MACHINES

Definition

A Moore machine associates output exclusively with states.

Six-Tuple Definition

$(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

Where:

Q = Set of states

Σ = Input alphabet

Δ = Output alphabet

q_0 = Initial state

$\delta = Q \times \Sigma \rightarrow Q$

$\lambda = Q \rightarrow \Delta$

Operational Characteristics

Output Length:

If input length = n

Output length = $n + 1$

Reason:

Initial state produces output before input processing.

Response to ϵ :

Moore machines produce output for ϵ (output of initial state).

State Dependency:

Output depends only on the current state.

DETAILED ANALYSIS OF MEALY MACHINES

Definition

A Mealy machine associates output with transitions (state-input pair).

Six-Tuple Definition

$(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

Where:

$$\delta = Q \times \Sigma \rightarrow Q$$

$$\lambda = Q \times \Sigma \rightarrow \Delta$$

Operational Characteristics

Output Length:

If input length = n

Output length = n

Response to ϵ :

No output is produced for ϵ .

Combined Dependency:

Output depends on both current state and current input symbol.

COMPARATIVE SUMMARY: MOORE VS MEALY

Feature	Moore Machine	Mealy Machine
Output Association	State-based	Transition-based
Output Function λ	$\lambda: Q \rightarrow \Delta$	$\lambda: Q \times \Sigma \rightarrow \Delta$

Output Length (input n)	$n + 1$	n
Response to ϵ	Yes	No
State Count in Conversion	No change when converting to Mealy	May increase when converting to Moore

CONVERSION AND EQUIVALENCE

Moore and Mealy machines have equal computational power. Each can simulate the other.

MOORE TO MEALY CONVERSION

This conversion does not increase the number of states.

Procedure:

1. Remove state outputs.
2. Transfer each state's output to all incoming transitions of that state.

If state Q_i has output 0 in Moore form, then every transition entering Q_i in Mealy form is labeled with output 0.

MEALY TO MOORE CONVERSION

This conversion may increase the number of states.

State Splitting:

If a state receives transitions with different outputs, it must be split into separate states, each representing one output.

Maximum State Bound:

$$\text{Total Moore States} \leq |Q| \times |\Delta|$$

Where:

$|Q|$ = Number of Mealy states

$|\Delta|$ = Number of output symbols

